Creator Defense

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected creator-defence record.

Explain the architecture, trace rates, audit overclaims and design a revealing question.

Keep one habit: Keep source versions, outcome units, denominators, failure evidence and unmeasured claims explicit.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Capstone Evaluate and Improve. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Capstone Design. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Bias and Representation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Capstone Evaluate and Improve

Capstone Design

Bias and Representation

W1 · Explain an evidence-backed defence

> Connect architecture, corpus, baseline, results and limitations.

Your evidence record

- W2 review: 9/10 retrieved-record relevance, 7/10 supported answers and zero recorded actions in one read-only run; no speed evidence. Main D1 per-question retrieval 12/15=80%, support 9/15=60%, separate fallback 4/5=80%. Individual failures, timing and complete reproduction details absent. D2 club note Cedar meets Tuesday, leader absent. Supplied practice counts are not a student's actual project evidence.

- Reference for this exercise: open this week's student lesson, "Creator Defense", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the relevance/support distinction

> Do not call retrieval success full answer correctness.

Your evidence record

- W2 review: 9/10 retrieved-record relevance, 7/10 supported answers and zero recorded actions in one read-only run; no speed evidence. Main D1 per-question retrieval 12/15=80%, support 9/15=60%, separate fallback 4/5=80%. Individual failures, timing and complete reproduction details absent. D2 club note Cedar meets Tuesday, leader absent. Supplied practice counts are not a student's actual project evidence.

- System reports 9/10 retrieved relevant records, 7/10 supported answers and zero actions in a read-only run. Presenter claims 90% fully correct and faster than baseline.

  Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Audit a universal-success claim

> A limited run leaves untested conditions.

Your evidence record

- W2 review: 9/10 retrieved-record relevance, 7/10 supported answers and zero recorded actions in one read-only run; no speed evidence. Main D1 per-question retrieval 12/15=80%, support 9/15=60%, separate fallback 4/5=80%. Individual failures, timing and complete reproduction details absent. D2 club note Cedar meets Tuesday, leader absent. Supplied practice counts are not a student's actual project evidence.

- A report omits this qualification: A **defense** should preserve negative **evidence** and uncertainty rather than treating polished presentation as proof.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Design a revealing defence challenge

> State a complete case, expected evidence need and honest missing result.

Your evidence record

- W2 review: 9/10 retrieved-record relevance, 7/10 supported answers and zero recorded actions in one read-only run; no speed evidence. Main D1 per-question retrieval 12/15=80%, support 9/15=60%, separate fallback 4/5=80%. Individual failures, timing and complete reproduction details absent. D2 club note Cedar meets Tuesday, leader absent. Supplied practice counts are not a student’s actual project evidence.

- Reference for this exercise: open this week’s student lesson, “Creator Defense”, at “See how it works” and “Follow a worked example”. Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

W4 · Defend the fresh outcome figures independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Use only the independent rates' stated counts and missing-latency fact. The outcome units and denominators must be named separately. No measured speed, energy comparison or universal correctness result is supplied.

A **defense** has retrieval 8/10, support 6/10, fallback 3/4 and no measured latency. State the three rates and reject one unsupported claim.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Final product card documents purpose, system map, sources, evals, limits, risks, and human oversight.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Trace the worked case for Generative systems **defense**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Generative systems **defense**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Your assistant passes 9/12 final questions, including all six direct facts but only 3/6 ambiguous or absent-**evidence** cases. Report results and argue for a focused next step.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Ten successful demos lead to a never hallucinates claim.

Write a defensible conclusion.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

An audience asks whether your assistant is always correct. Give an **evidence**-based **defense** using results and a limit.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 36 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Your assistant passes 9/12 final questions, including all six direct facts but only 3/6 ambiguous or absent-**evidence** cases. Report results and argue for a focused next step.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A **defense** has retrieval 8/10, support 6/10, fallback 3/4 and no measured latency. State the three rates and reject one unsupported claim.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

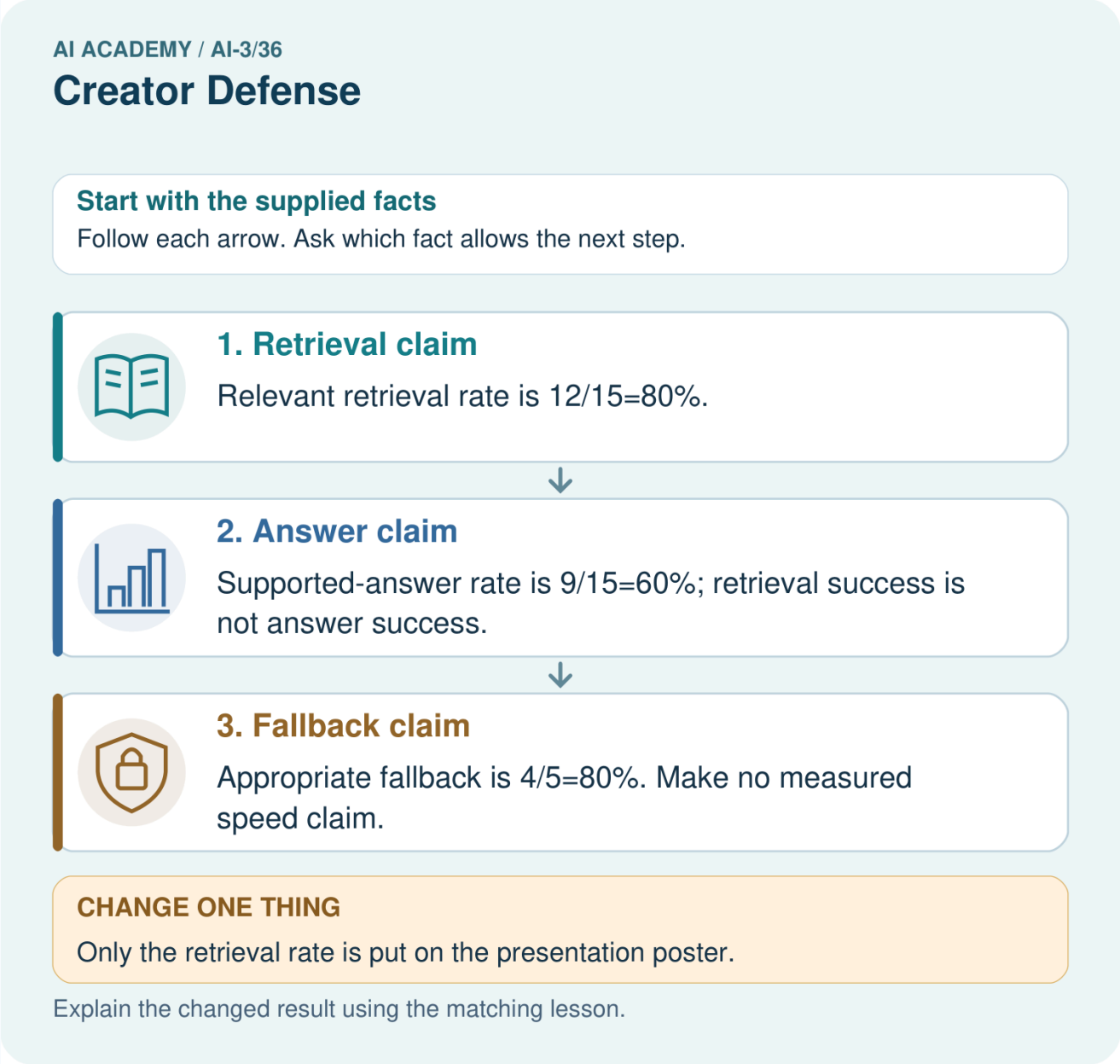
> Keep the explanation and the support you used.

Use this when helpful

Defend a study-buddy project using separate claims and matching **evidence**.

Set A: 15 answerable questions, with 12 relevant retrievals and 9 supported answers. Separate set B: 5 missing-source questions, with 4 appropriate fallbacks. No speed measurements exist.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

Which numbers would you display to make the **defense** honest? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Verified Study Buddy

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For your Verified Study Buddy, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- Defend a study-buddy project using separate claims and matching **evidence**.
- Which labelled step matches this detail? Appropriate fallback is 4/5=**80%**. Make no measured speed claim.
- 1. Fallback claim

- 2. Answer claim
- 3. Retrieval claim
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: Fallback claim Appropriate fallback is 4/5=**80%**. Make no measured speed claim. Read the picture in order. Appropriate fallback is 4/5=**80%**.

  Make no measured speed claim. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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> | |
> |---|
> | |
> | |
> | |

▼ **Check the first explanation**

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: Only the retrieval rate is put on the presentation poster. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.

- 2. The poster omits the lower supported-answer rate and cannot imply **80%** fully correct answers.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.
- After your attempt, compare with the supported explanation: The poster omits the lower supported-answer rate and cannot imply **80%** fully correct answers.

  Retrieving useful **evidence** is an intermediate step, not proof that the final answer uses it correctly.

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

Repair a missing building block

Repair and save your work

Review Capstone Evaluate and Improve (ai-3/35). Goal: Explain the **mechanism** and terminology of capstone evaluation and revision. Short practice: May the team call the repaired validation cases an untouched final test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Capstone Design (ai-3/33). Goal: Explain the **mechanism** and terminology of capstone requirements and corpus. Short practice: What should the project plan record before the new flower test?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Bias and Representation (ai-3/30). Goal: Explain the **mechanism** and terminology of representation and bias evaluation. Short practice: Explain why comparing six correct answers in each group would mislead.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

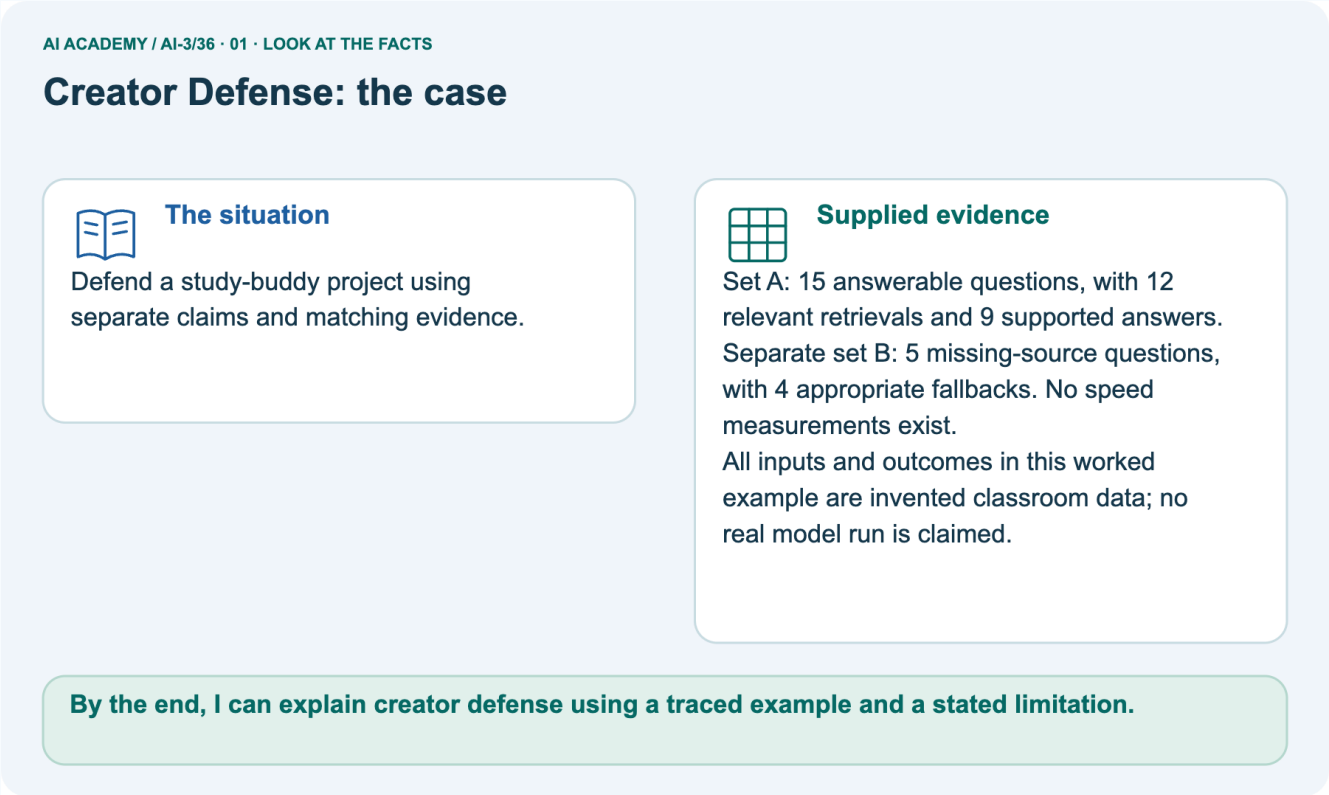
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Creator Defense: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

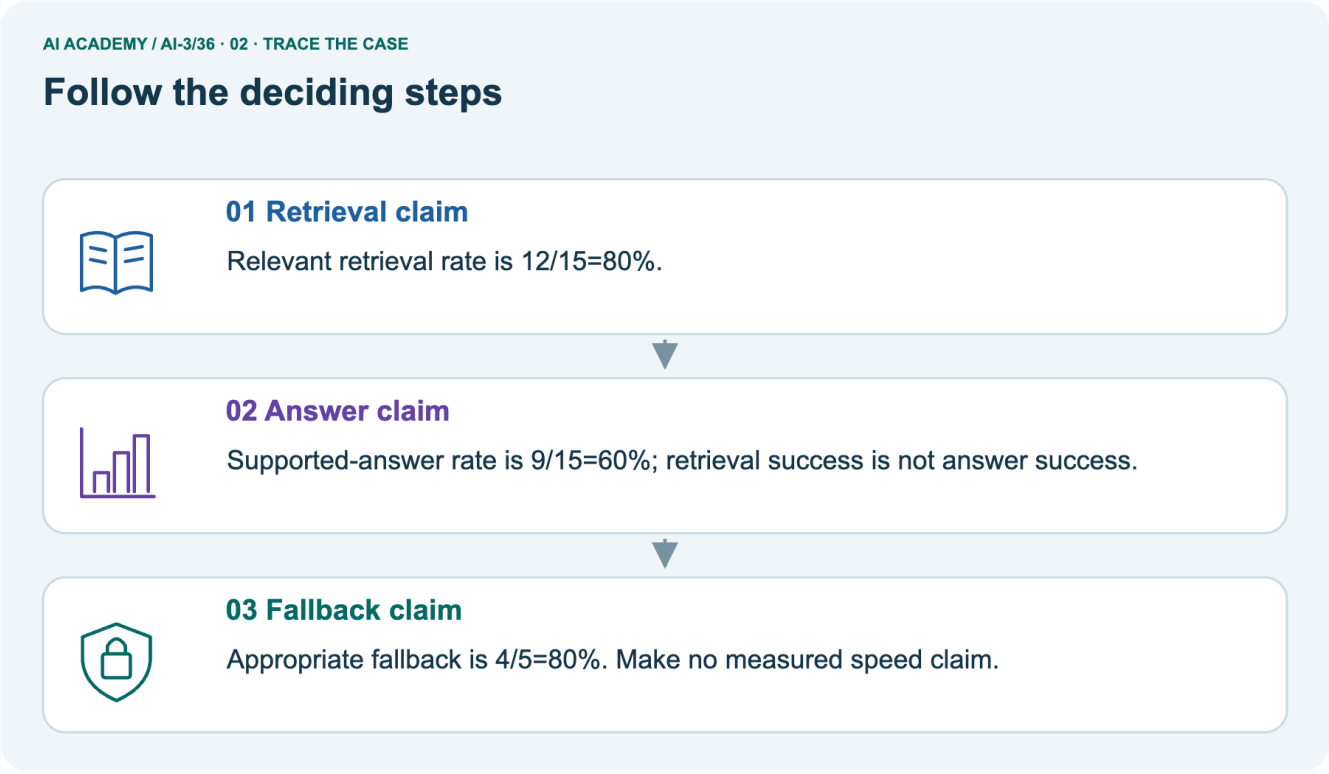
Defend a study-buddy project using separate claims and matching **evidence**. Supplied facts: Set A: 15 answerable questions, with 12 relevant retrievals and 9 supported answers.

Separate set B: 5 missing-source questions, with 4 appropriate fallbacks. No speed measurements exist. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

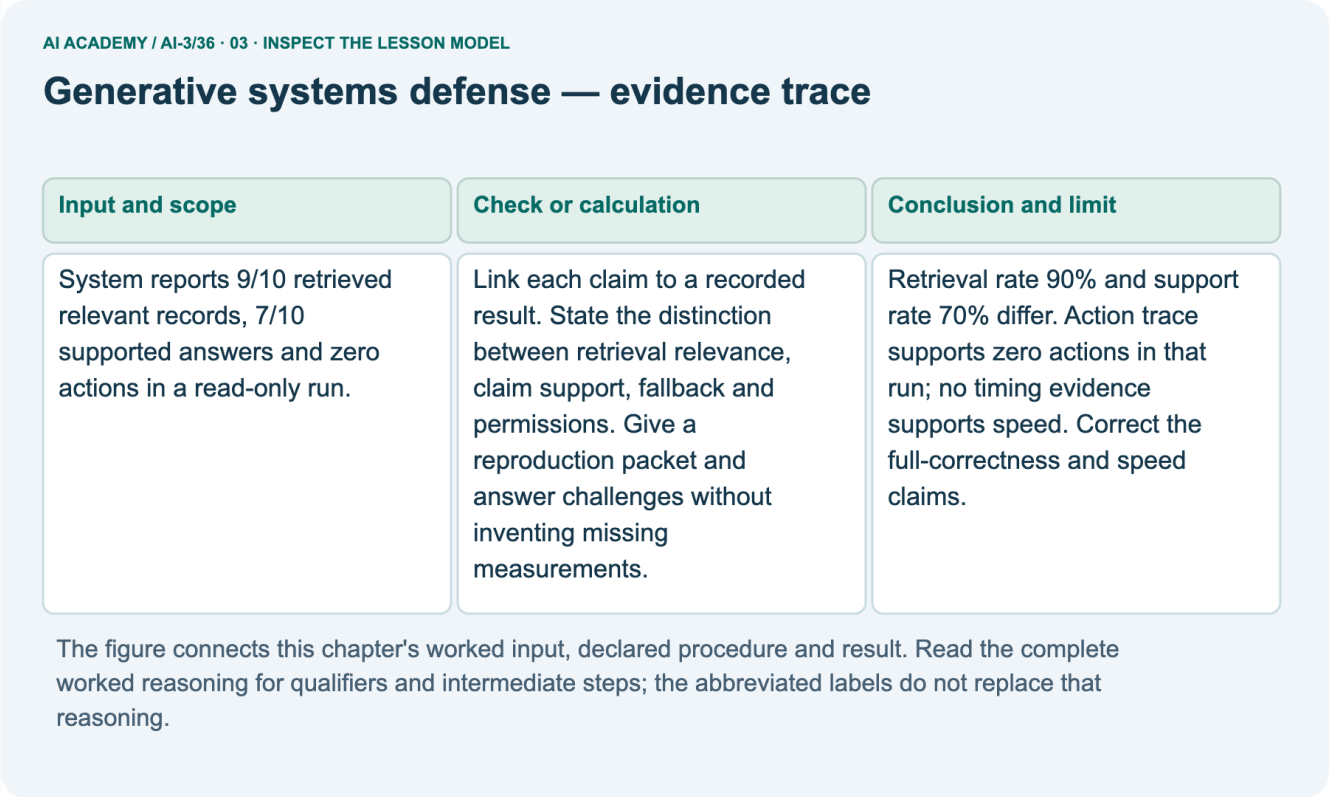
Retrieval claim: Relevant retrieval rate is 12/15=**80%**. → Answer claim: Supported-answer rate is 9/15=**60%**; retrieval success is not answer success. → Fallback claim: Appropriate fallback is 4/5=**80%**.

Make no measured speed claim.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Generative systems defense — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Generative systems **defense** — **evidence** trace. System reports 9/10 retrieved relevant records, 7/10 supported answers and zero actions in a read-only run. | Link each claim to a recorded result.

State the distinction between retrieval relevance, claim support, fallback and permissions.

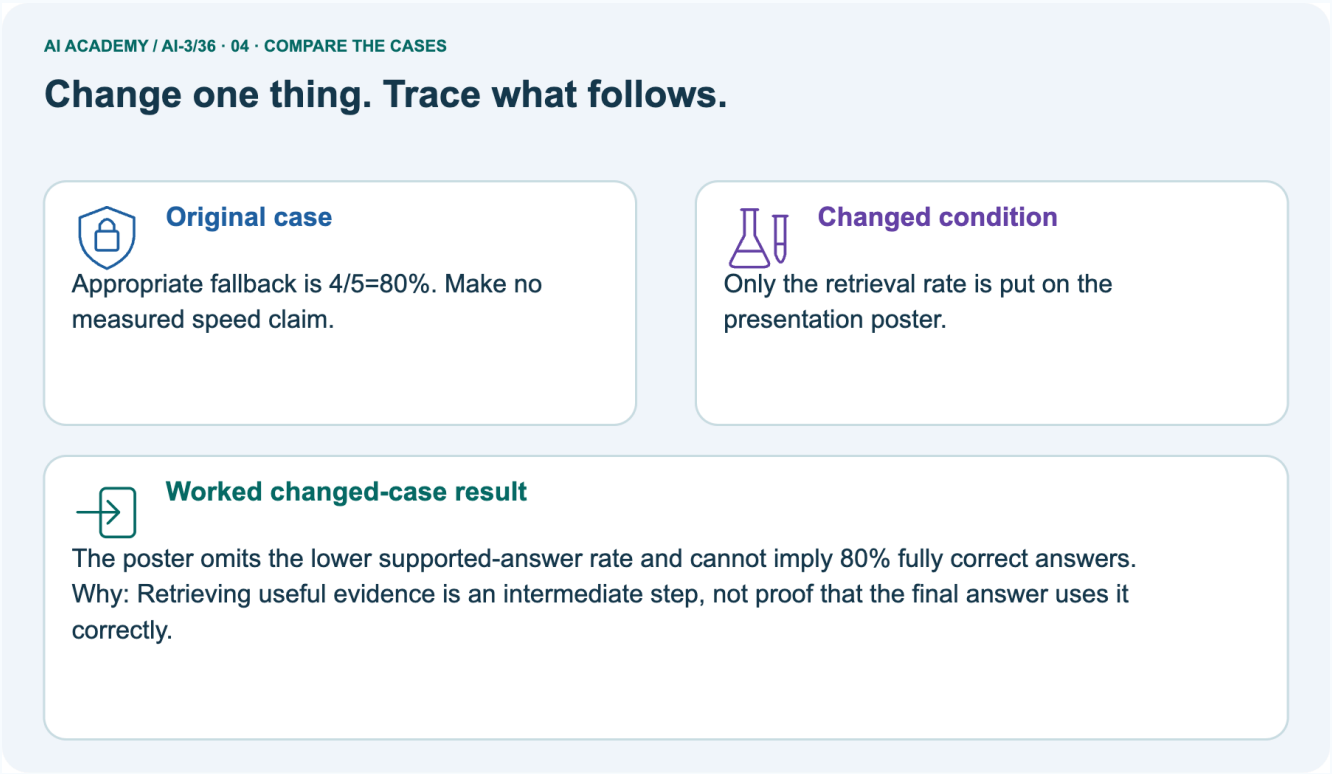
Give a **reproduction packet** and answer challenges without inventing missing measurements. | Retrieval rate 90% and support rate **70%** differ. Action trace supports zero actions in that run; no timing **evidence** supports speed.

Correct the full-correctness and speed claims.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

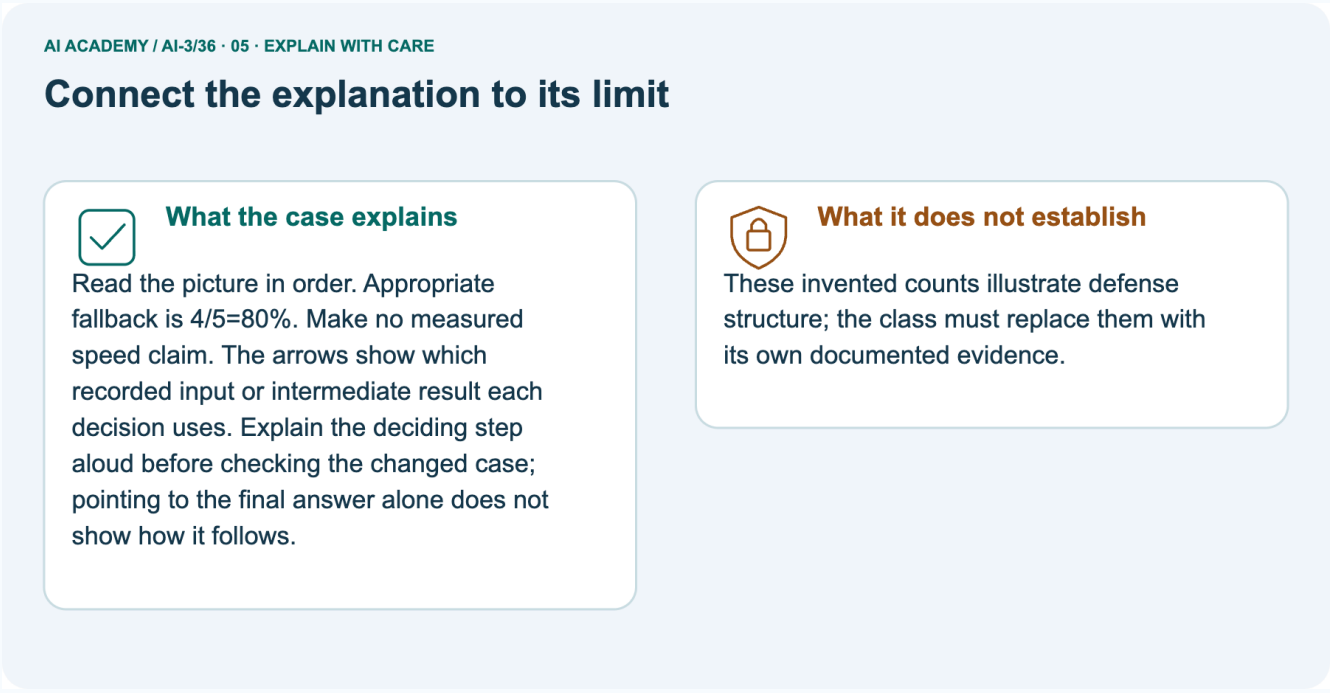
Original case: Appropriate fallback is 4/5=**80%**. Make no measured speed claim. Changed condition: Only the retrieval rate is put on the presentation poster.

Expected result: The poster omits the lower supported-answer rate and cannot imply **80%** fully correct answers. Reason: Retrieving useful **evidence** is an intermediate step, not proof that the final answer uses it correctly.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Appropriate fallback is 4/5=**80%**. Make no measured speed claim. The arrows show which recorded **input** or intermediate result each decision uses.

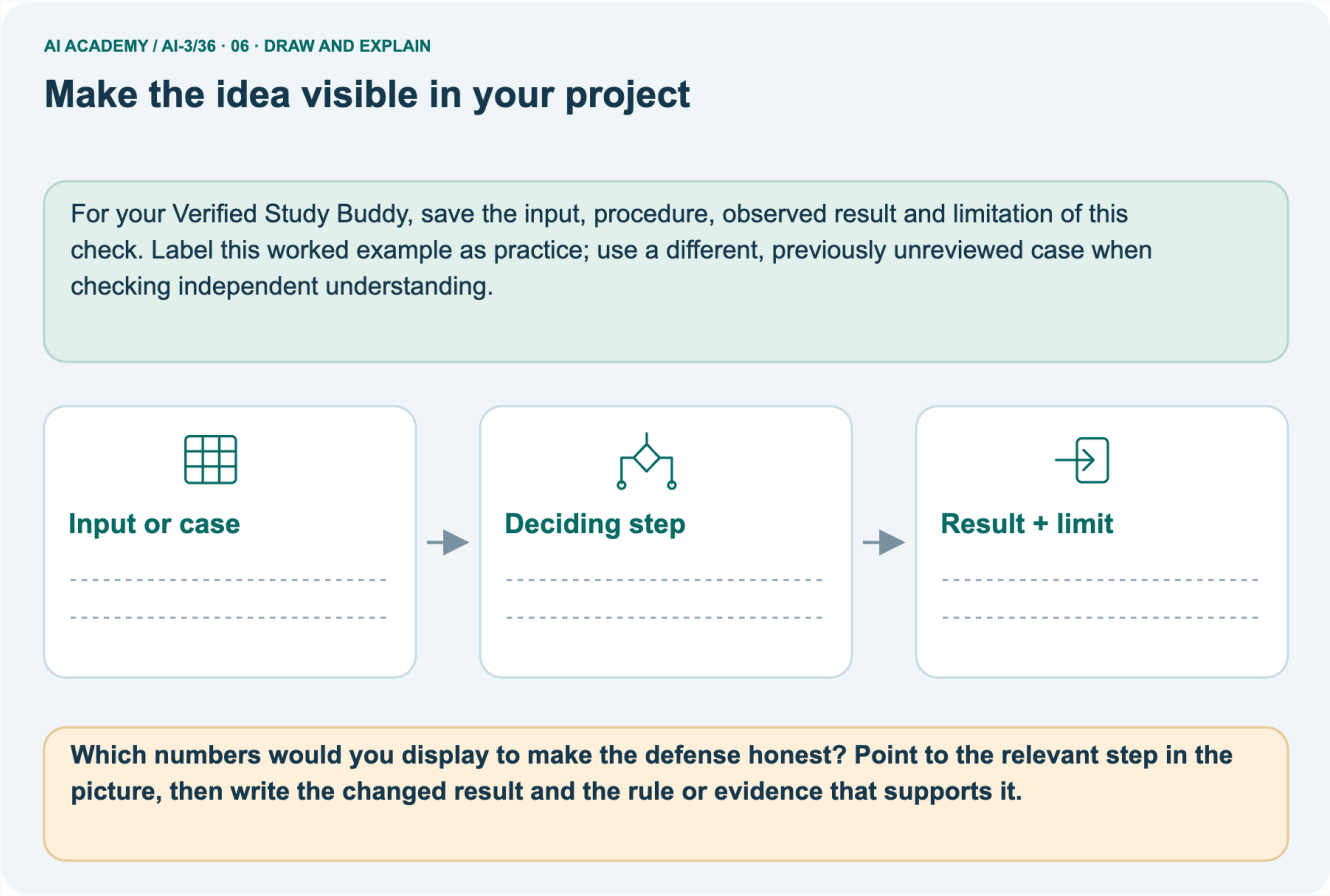
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: These invented counts illustrate **defense** structure; the class must replace them with its own documented **evidence**.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and **limitation** of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which numbers would you display to make the **defense** honest? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

> **My picture explanation, deciding evidence and one limit**

▼ **Earlier wording, alternatives and picture transcripts**

01. Creator Defense: the case

02. Follow the deciding steps

03. Generative systems defense — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ **Optional interactive practice and completed work**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.